Integration by Parts

Questions 1–30

Question 1. Find

a) $\int xe^x \, dx$

b) $\int (x-1)e^x \, dx$

c) $\int (2x+5)e^x \, dx$

Question 2. Evaluate the following definite integral:

$$\int_{-\frac{1}{3}}^{\frac{1}{3}} xe^{3x} \, dx$$

Question 3. Evaluate the following definite integral:

$$\int_{-1}^{0} \frac{x+1}{3e^x}\,dx$$

Question 4. The curve y has gradient function

$$\frac{dy}{dx} = 2x\sqrt{e^x}$$

and passes through the point $(\ln 4, 2)$. Show that y can be written in the form

$$y = a(x-2)\sqrt{e^x} + b\ln 4 + c,$$

where a, b and c are integers.

Question 5. Find

a) $\int x \sin x \, dx$

b) $\int \left(x + \frac{\pi}{3}\right) \sin x \, dx$

Question 6. Evaluate the following definite integral:

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x \sin 2x \, dx$$

Question 7. Find

a) $\int x \cos x \, dx$

b) $\int \left(x - \frac{\pi}{6}\right) \cos x \, dx$

Question 8. Evaluate the following definite integral:

$$\int_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} \left(x - \frac{2\pi}{3}\right) \cos x \, dx$$

Question 9. Evaluate the following definite integral:

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \left(2x - \frac{\pi}{4}\right) \sin 2x \, dx$$

Question 10. Evaluate the following definite integral:

$$\int_{-\frac{\pi}{8}}^{\frac{\pi}{8}} \left(x + \frac{\pi}{8}\right) \cos 4x \, dx$$

Question 11. The curve y has gradient function

$$\frac{dy}{dx} = 64x \sin 8x$$

and passes through the point $\left(\frac{\pi}{16}, 1\right)$. Show that y can be written in the form

$$y = a \sin 8x + bx \cos 8x,$$

where a and b are integers.

Question 12. The curve y has gradient function

$$\frac{dy}{dx} = 6x \cos \frac{1}{3}x$$

and passes through the point $(\pi, -1)$. Show that y can be written in the form

$$y = ax \sin \frac{1}{3}x + b \cos \frac{1}{3}x + p\sqrt{3}\pi + q,$$

where a, b, p and q are integers.

Question 13. Find

a) $\int x \ln x \, dx$

b) $\int x^2 \ln x \, dx$

c) $\int (x+4) \ln x \, dx$

d) $\int (x^3 + 3x) \ln x \, dx$

Question 14. Evaluate the following definite integral:

$$\int_1^2 \frac{\ln 8x}{x^4} \, dx$$

Question 15. The curve y has gradient function

$$\frac{dy}{dx} = 4x \ln \frac{1}{2}x$$

and passes through the point $(2e, 1)$. Show that y can be written in the form

$$y = ax^2 + bx^2 \ln \frac{1}{2}x + pe^2 + q,$$

where a, b, p and q are integers.

Question 16. Find

a) $\int x2^x \, dx$

b) $\int (3x + 9)3^x \, dx$

Question 17. Evaluate the following definite integral:

$$\int_0^1 (x+2)2^x \, dx$$

Question 18. Evaluate the following definite integral:

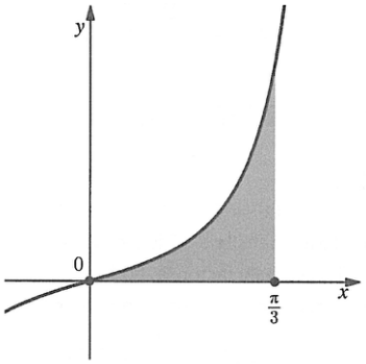
$$\int_1^2 10x5^{x-1} \, dx$$

Question 19. The curve y has gradient function

$$\frac{dy}{dx} = \frac{x}{4^x}$$

and passes through the point $(\ln 4, 1)$. Write y in terms of x.

Question 20. The diagram below shows the curve $y = x\sec^2 x$. Show that the exact shaded area is $\frac{\pi}{\sqrt{3}} - \ln 2$.



Question 21. Evaluate the following definite integral:

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 18x \operatorname{cosec}^2 x \, dx$$

Question 22. Find

a) $\int \ln x \, dx$

b) $\int (\ln x)^2 \, dx$

Question 23. Prerequisite knowledge for Q23: Integration by Substitution
Evaluate the following definite integral:

$$\int_3^{e^2+2} \ln(x-2)\,dx$$

Question 24. Integration by Parts Twice Find

a) $\int x^2 e^x\,dx$

b) $\int \frac{x^2}{e^x}\,dx$

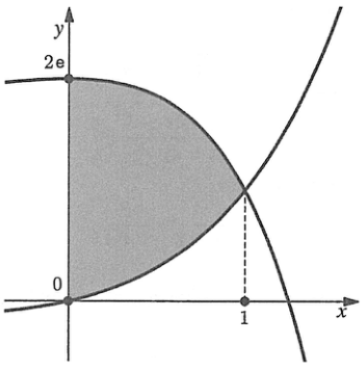
c) $\int x^2 \sin x\,dx$

d) $\int x^2 2^x\,dx$

Question 25. Evaluate the following definite integral:

$$\int_0^{\frac{1}{4}} 20x^2e^{-4x}\,dx$$

Question 26. The diagram below shows the curves $y = xe^x$ and $y = 2e - x^2e^x$. Find the exact shaded area between the two curves.



Question 27. Evaluate the following definite integral:

$$\int_1^e (\ln x)^3 \, dx$$

Question 28. Cyclic Integration Find

a) $\int e^x \sin x \, dx$

b) $\int e^x \cos x \, dx$

Question 29. Evaluate the following definite integral:

$$\int_0^{\frac{\pi}{2}} e^{5x} \sin 2x \, dx$$

Question 30. The diagram below shows the curve $y = \ln(2x - k)$. It is given that the curve passes through the origin, O, and the point A with x-coordinate 1. The normal to the curve at A intersects the x-axis at B. Find the exact shaded area of the shape OAB.

